

Answer key for the sample papers

Generative AI and Large Language Models

GenAI crash course

2026-10-10

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Numeric answers were computed by `site_src/assessment.py` using the course modules. Short answers are model answers; accept any answer that says the same thing.

Mid-semester paper (Units 1 to 5): answers

1. (2, Unit 1) Reconstruction term: makes the decoder rebuild the input from the latent code. KL term: keeps the encoder's latent distribution close to the prior (a standard normal), so that sampling from the prior gives valid codes.

2. (2, Unit 1) C.

3. (4, Unit 1) An autoregressive model factorises $p(x) = \prod p(x_t | x_{<t})$ exactly. A VAE needs the integral over the latent variable, which is intractable, so it maximises the evidence lower bound (ELBO) instead.

4. (1, Unit 2) B.

5. (2, Unit 2) $12 \times 768^2 = 7,077,888$.

6. (2, Unit 2) $4 \times 2,048^2 \times 1,024 = 17,179,869,184$ (about 17.2 GFLOPs per layer).

7. (3, Unit 2) If query and key entries have unit variance, the dot product has variance d_k . Without scaling, large d_k pushes the softmax into a saturated region with tiny gradients. Dividing by $\sqrt{d_k}$ restores unit variance.

8. (2, Unit 2) C.

9. (4, Unit 2) The causal mask lets position t see only positions $\leq t$, so the model is trained to predict the next token and can be sampled one token at a time. BERT sees both sides during training (masked-token prediction), so it has no left-to-right factorisation to sample from.

10. (1, Unit 3) $T = 1$: (0.665, 0.245, 0.090). $T = 0.5$: (0.867, 0.117, 0.016). Lower temperature sharpens the distribution.

11. (2, Unit 3) (1) Start with characters as tokens. (2) Count all adjacent pairs and merge the most frequent pair into a new token. (3) Repeat for the chosen number of merges; the ordered list of merges is the tokenizer.
12. (3, Unit 3) $N = \sqrt{C/120} = 2.887e+10$ parameters (about 28.9 billion); $D = 20 N = 5.774e+11$ tokens (about 577 billion).
13. (4, Unit 3) $L = 2.184$.
14. (2, Unit 4) $P(\text{at least 3 of 5 correct}) = 0.8369$.
15. (2, Unit 4) The model may return malformed or wrong-type output. Validate against a schema; on failure, retry with the error message in the prompt, up to a limit, then fail loudly.
16. (4, Unit 4) The oldest messages are dropped first, which can remove the user's name, constraints or few-shot examples. Pin the messages that must stay (system message, key facts) and trim the rest.
17. (2, Unit 5) $1/(60+1) + 1/(60+3) = 0.03227$.
18. (2, Unit 5) 4 chunks (computed with the course's `chunk_text`).
19. (2, Unit 5) Retrieval misses the relevant chunk (vocabulary mismatch, bad chunking); the context is too long or noisy so the model ignores the right passage; the model answers from memory or invents text that the context does not support; the documents are outdated.
20. (4, Unit 5) Retrieval is fine; the loss is in selection or generation (right chunk, wrong sentence or a misreading). Evaluate stage by stage: the share of questions whose top chunk contains the fact, and the share of those answered correctly (Unit 11).

End-semester paper (Units 1 to 11, weighted to Units 6 to 11): answers

1. (2, Unit 1) Reconstruction term: makes the decoder rebuild the input from the latent code. KL term: keeps the encoder's latent distribution close to the prior (a standard normal), so that sampling from the prior gives valid codes.
2. (2, Unit 1) C.
3. (4, Unit 1) An autoregressive model factorises $p(x) = \prod p(x_t | x_{<t})$ exactly. A VAE needs the integral over the latent variable, which is intractable, so it maximises the evidence lower bound (ELBO) instead.
4. (2, Unit 2) $12 \times 768^2 = 7,077,888$.
5. (2, Unit 2) $4 \times 2,048^2 \times 1,024 = 17,179,869,184$ (about 17.2 GFLOPs per layer).
6. (3, Unit 2) If query and key entries have unit variance, the dot product has variance d_k . Without scaling, large d_k pushes the softmax into a saturated region with tiny gradients. Dividing by $\sqrt{d_k}$ restores unit variance.
7. (4, Unit 2) The causal mask lets position t see only positions $\leq t$, so the model is trained to predict the next token and can be sampled one token at a time. BERT sees both sides during training (masked-token prediction), so it has no left-to-right factorisation to sample from.
8. (1, Unit 3) $T = 1$: (0.665, 0.245, 0.090). $T = 0.5$: (0.867, 0.117, 0.016). Lower temperature sharpens the distribution.
9. (3, Unit 3) $N = \sqrt{C/120} = 2.887e+10$ parameters (about 28.9 billion); $D = 20 N = 5.774e+11$ tokens (about 577 billion).

10. (2, Unit 4) $P(\text{at least 3 of 5 correct}) = 0.8369$.
11. (4, Unit 4) The oldest messages are dropped first, which can remove the user's name, constraints or few-shot examples. Pin the messages that must stay (system message, key facts) and trim the rest.
12. (2, Unit 5) $1/(60+1) + 1/(60+3) = 0.03227$.
13. (2, Unit 5) 4 chunks (computed with the course's `chunk_text`).
14. (4, Unit 5) Retrieval is fine; the loss is in selection or generation (right chunk, wrong sentence or a misreading). Evaluate stage by stage: the share of questions whose top chunk contains the fact, and the share of those answered correctly (Unit 11).
15. (1, Unit 6) C.
16. (2, Unit 6) $r(d_{\text{out}} + d_{\text{in}}) = 8 \times 8192 = 65,536$; the matrix has 16,777,216 entries, so the fraction is 0.39%.
17. (2, Unit 6) 16 bits = 2 bytes: 14 GB. 4 bits = 0.5 bytes: 3.5 GB.
18. (3, Unit 6) Fine-tuning on a narrow task overwrites earlier abilities. Reduce it with a low learning rate or few steps, LoRA (the base weights stay frozen), or mixing in general data.
19. (4, Unit 6) The prompt is given context, not a target: training to reproduce it wastes capacity and lets prompt text dominate the gradient. The model should learn to produce the response given the prompt.
20. (1, Unit 7) 224×224 with 16×16 patches: 196 tokens. 336×336 with 14×14 patches: 576 tokens.
21. (2, Unit 7) D. A contrastive loss over a batch.
22. (3, Unit 7) $x_t = \sqrt{a_{\text{bar}}_t} x_0 + \sqrt{1 - a_{\text{bar}}_t} \text{eps}$ with $\text{eps} \sim N(0, I)$. Any noise level can be produced in one step from x_0 , so training samples random t without simulating the chain.
23. (4, Unit 7) The noise estimate is pushed away from the unconditional prediction toward the conditional one: $\text{eps}_{\text{hat}} = \text{eps}_{\text{uncond}} + (1 + w)(\text{eps}_{\text{cond}} - \text{eps}_{\text{uncond}})$. A larger w follows the prompt more closely but reduces diversity and can distort the image.
24. (1, Unit 8) D.
25. (2, Unit 8) $0.95^{10} = 0.599$; $0.99^{10} = 0.904$.
26. (3, Unit 8) Treat tool output as data, never as instructions; restrict the agent to an allowlist of tools and require confirmation for sensitive actions (such as sending email); filter or isolate untrusted text.
27. (4, Unit 8) Fully connected: $n(n-1) = 20$. Hub: $2(n-1) = 8$.
28. (1, Unit 9) C. It does not measure truth.
29. (2, Unit 9) $\kappa = 0.583$ (observed agreement 0.80, chance agreement 0.52).
30. (2, Unit 9) $[0.70, 0.97]$.
31. (3, Unit 9) Position bias: the judge prefers the answer shown first. Ask twice with the order swapped and average, or discard inconsistent verdicts.
32. (4, Unit 9) $b = 1/0.5 = 2.0$; standard deviation = $\sqrt{2} b = 2.83$.
33. (1, Unit 10) D.
34. (2, Unit 10) $2 \times 32 \times 8 \times 128 \times 2 = 131,072$ bytes per token; for 2000 tokens 262,144,000 bytes

(about 262 MB).

35. (3, Unit 10) $2 / (1000 \times 0.5 \times 3600) \times 10^6 = 1.11$ dollars.

36. (2, Unit 10) The mean hides the slow tail that users notice; the 95th percentile says how slow the slowest 5% of requests are, which is what a service-level objective should bound.

37. (2, Unit 11) User, data, minimum viable version, the one metric, main risk. Without a metric you cannot tell when the project works; writing it first stops you from choosing the metric after seeing the result.

38. (2, Unit 11) $[0.71, 0.93]$.

39. (3, Unit 11) $\pi > c_H / (c_H + c_F) = 9/10 = 0.90$.

40. (4, Unit 11) The threshold is tuned to those questions, so the result is optimistic. Split into a development half (choose settings) and a test half (report once), and give confidence intervals because small sets are noisy.